

Week 3 Introduction

Mathematics for Multidimensional Data

✓ **Video:** 3.1 Multidimensional Data Points and Features
2 min

✓ **Reading:** Multidimensional Data Points and Features Recap
10 min

✓ **Practice Quiz:** Multidimensional Data Points and Features – Review Information
1 question

▶ **Video:** 3.2 Multidimensional Mean
2 min

✓ **Reading:** Multidimensional Mean Recap
10 min

✓ **Practice Quiz:** Multidimensional Mean – Review Information
1 question

▶ **Video:** 3.3 Dispersion: Multidimensional Variables
3 min

✓ **Reading:** Multidimensional Variables Recap
10 min

📖 **Practice Quiz:** Dispersion: Multidimensional Variables – Review Information
3 questions

▶ **Video:** 3.4 Distance Metrics
5 min

📖 **Reading:** Distance Metrics Recap
10 min

📖 **Practice Quiz:** Distance Metrics – Review Information
4 questions



Multidimensional Variables Recap

Variance and Standard Deviation

Here are our two formulas:

$$\sigma_j^2 = \frac{\sum_{i=1}^N (x_{ij} - \mu_j)^2}{N}, \text{ for the } \mathbf{population}, \text{ and}$$

$$s_j^2 = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}{(n-1)} = \frac{1}{(n-1)} \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2 \text{ for a } \mathbf{sample}.$$

Let me unpack this a bit:

You remember the concept of **deviation**, which tells you how *distant* each variable is from the mean (remember that you can have two sets of numbers with the same mean, one with numbers very spread out, the other with all of them very close to the mean), and calculated like so: $(x_i - \mu)$ for the population, $(x_i - \bar{x})$ for a sample.

When we wanted to calculate the average of all our deviations, which will give us the **variance** and then the **standard deviation**, we saw that they all summed up to zero, and so we squared them to cancel that, then divided by the infamous $(n-1)$ (I'll focus on the sample case from now on), using the following equations:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{(n-1)}, \text{ the } \mathbf{variance}, \text{ and}$$

$$s = \sqrt{s^2} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{(n-1)}}, \text{ the } \mathbf{standard deviation}.$$

As you can see, these are the equations above, only without the $i = 1$ and the n (like so: $\sum_{i=1}^n$) specifying that we are working on n observations!

Covariance

The concept of covariance expresses how much variables are linked, or vary, together. Intuitively, you can imagine yourself